Reformulating Koide's Mass Relation Using Inverse Compton Radii in Wave Theory

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Abstract

Koide's empirical mass relation for leptons exhibits remarkable numerical precision but lacks a theoretical foundation. In this paper, we re-express Koide's formula in terms of inverse reduced Compton radii, assuming a curvature-based mass definition arising from the geometric framework of Wave Theory. The resulting form reveals a purely geometric structure that supports the view of mass as emerging from universal curvature modulation.

1 Introduction

Koide's formula has long intrigued physicists by predicting the mass ratios of the three charged leptons (electron, muon, tau) with astonishing precision:

$$Q = \frac{m_1 + m_2 + m_3}{\left(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3}\right)^2} \approx \frac{2}{3}. \tag{1}$$

Despite its accuracy—agreeing with experimental lepton masses to many significant digits—the formula has no accepted theoretical origin within the Standard Model of particle physics. Its structure, involving square roots and a fixed ratio of $\frac{2}{3}$, suggests an underlying symmetry or geometric principle yet to be uncovered.

Over the decades, numerous attempts have been made to explain the formula's origin using flavor symmetries, grand unification theories, or speculative group-theoretic frameworks. However, none have provided a definitive derivation from first principles.

Wave Theory offers a geometric mass model in which mass arises from curvature modulation embedded within a 3-sphere universe. In this framework, a particle's mass is inversely related to its reduced Compton radius. This allows Koide's formula to be reformulated not in terms of masses, but in terms of inverse radii—yielding a clean geometric interpretation of the mass hierarchy observed in charged leptons.

2 Wave Theory Mass Definition

In Wave Theory, mass is not treated as a fundamental parameter but as a manifestation of geometric curvature embedded in spacetime. Specifically, the mass of a particle is defined as:

$$m_i = \frac{\pi L_p^2 R}{r_i}, \tag{2}$$

where L_p is the Planck length, R is the current radius of the observable 3-sphere universe, and r_i is the reduced Compton radius of particle i.

This relation reflects the idea that mass arises from a standing wave curvature confined to a region of radius r_i. The factor $\pi L_p^2 R$ represents a universal curvature energy scale, while the inverse dependence on r_i captures the localization of that curvature. Smaller Compton radii correspond to more tightly curved standing wave configurations and therefore to higher mass.

In the unit system of Wave Theory, mass is expressed dimensionally in terms of length squared, $[m] = L^2$, reflecting its role as a curvature density rather than a fundamental substance. The quantity $\pi L_p^2 R$ thus encodes a global surface curvature potential, and the inverse Compton radius acts as a measure of localized compression against that universal scale.

Interestingly, this expression can also be decomposed as:

$$m_i = (\pi L_p R) \cdot \left(\frac{L_p}{r_i} \right)$$

where $\pi L_p R$ has the units of mass and may be interpreted as a **Planck-scale mass envelope**, while L_p/r_i represents the **relative curvature compression** of the wave at scale r_i. In this view, the particle's mass emerges as the modulation of a Planck mass by the wave's confinement factor.

Importantly, this formula allows for direct computation of a particle's mass given its reduced Compton radius, without requiring fine-tuning or empirical mass fitting. This stands in contrast to conventional models where mass is introduced as an external parameter. This predictive relation is numerically confirmed in Section 5, where measured Compton radii of charged leptons are shown to reproduce Koide's mass ratio with high precision.

Moreover, the formula implies that as the universe expands and R increases, the global energy scale governing mass decreases—embedding the mass spectrum in the geometry of cosmic evolution.

3 Substitution into Koide

We now substitute this mass definition into the Koide formula.

Numerator:

$$m_1 + m_2 + m_3 = \pi L_p^2 R \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right) \tag{3}$$

Denominator:

$$\begin{aligned}
(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3})^2 &= \left(\sqrt{\frac{\pi L_p^2 R}{r_1}} + \sqrt{\frac{\pi L_p^2 R}{r_2}} + \sqrt{\frac{\pi L_p^2 R}{r_3}} \right)^2 \qquad (4) \\
&= \left(\sqrt{\pi L_p^2 R} \left(\frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}} + \frac{1}{\sqrt{r_3}} \right) \right)^2 \qquad (5) \\
&= \pi L_p^2 R \left(\sum_i \frac{1}{\sqrt{r_i}} \right)^2 \qquad (6)
\end{aligned}$$

4 Final Expression

Substituting numerator and denominator into the Koide formula:

$$\begin{aligned}
Q &= \frac{\pi L_p^2 R \sum_i \frac{1}{r_i}}{\pi L_p^2 R \left(\sum_i \frac{1}{\sqrt{r_i}} \right)^2} \qquad (7) \\
&= \frac{\sum_i \frac{1}{r_i}}{\left(\sum_i \frac{1}{\sqrt{r_i}} \right)^2} \qquad (8)
\end{aligned}$$

This leads to the final form:

$$Q = \frac{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}{\left(\frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}} + \frac{1}{\sqrt{r_3}}\right)^2}$$

5 Numerical Evaluation for Leptons

The reduced Compton radius of a particle is defined as:

$$Q = \frac{\frac{1}{r_e} + \frac{1}{r_\mu} + \frac{1}{r_\tau}}{\left(\frac{1}{\sqrt{r_e}} + \frac{1}{\sqrt{r_\mu}} + \frac{1}{\sqrt{r_\tau}}\right)^2} \approx 0.66666049 \tag{9}$$

where $\hbar$ is the reduced Planck constant, m_i is the particle's rest mass, and c is the speed of light.

Unlike the full Compton wavelength $\lambda = h/mc$, the reduced form omits the 2π factor and corresponds to the natural radial extent of a quantum wavefunction in relativistic quantum mechanics. In Wave Theory, this reduced radius is interpreted as the curvature scale of a localized standing wave embedded within the 3-sphere universe. It is the scale at which the wave is anchored in time and space—serving as a geometric proxy for how tightly the wave is confined.

Using known reduced Compton radii for the charged leptons:

$$\begin{aligned} r_e &= 3.86159319 \times 10^{-13} \text{ m}, \\ r_\mu &= 1.86759452 \times 10^{-15} \text{ m}, \\ r_\tau &= 1.11053794 \times 10^{-16} \text{ m}, \end{aligned}$$

we substitute these values into the curvature-based form of the Koide formula:

$$Q = \frac{\frac{1}{r_e} + \frac{1}{r_\mu} + \frac{1}{r_\tau}}{\left(\frac{1}{\sqrt{r_e}} + \frac{1}{\sqrt{r_\mu}} + \frac{1}{\sqrt{r_\tau}}\right)^2} \approx 0.66666049. \tag{10}$$

This confirms that the Koide ratio, reformulated using inverse reduced Compton radii, gives a result consistent with $Q = \frac{2}{3}$ to at least eight decimal digits for the charged leptons.

6 Discussion

The result above expresses Koide's mass relation entirely in terms of inverse reduced Compton radii, assuming the Wave Theory definition of mass. This transformation shifts the perspective from mass as an intrinsic, unexplained parameter to a derived quantity governed by geometric curvature.

The structure of the final expression is notable: the numerator scales linearly with inverse radii, while the denominator involves a squared sum of inverse square roots. This asymmetry suggests a deeper coherence—possibly linked to phase relationships or wave interference among standing wave modes confined to the 3-sphere. The resulting value, $Q \approx \frac{2}{3}$, may reflect an equilibrium or resonance condition in curvature distribution among the charged leptons.

Attempts to extend Koide's formula to other fermions—such as quarks or neutrinos—have not yielded consistent or stable results. In the Wave Theory framework, this discrepancy may arise because only charged leptons form coherent curvature-bound states with clearly defined, stable Compton radii. Neutrinos, being near-massless and delocalized, and quarks, being confined, may not satisfy the same wave conditions required for Koide coherence.

This geometric reformulation opens several intriguing questions: Can other ratios—such as coupling constants or mixing angles—be re-expressed in purely curvature-based terms? Does the value $\frac{2}{3}$ emerge from a deeper symmetry of the 3-sphere geometry?

Further investigation may uncover whether Koide's relation is not just a numerical coincidence, but a manifestation of curvature resonance in a self-organizing quantum geometry.

7 Conclusion

Recasting Koide's mass relation using inverse reduced Compton radii within the framework of Wave Theory reveals a compelling geometric interpretation of the charged lepton mass hierarchy. By treating mass as a manifestation of curvature embedded in a dynamically expanding 3-sphere universe, the formula no longer relies on arbitrary constants or empirical tuning.

Instead, the relation emerges from universal geometric scaling—linking the size of a particle's Compton radius to its embedded wave curvature. The resulting expression matches the Koide ratio with remarkable precision, reinforcing the idea that lepton masses may reflect an underlying curvature resonance rather than unexplained numerical coincidence.

This approach demonstrates how purely geometric quantities—Planck length, cosmic radius, and Compton scale—can combine to reveal structural regularities in the particle spectrum. It provides evidence that mass relations may be governed by large-scale curvature constraints and opens the door to deeper unification through geometry.

Further extensions may explore whether similar curvature-based formulations apply to coupling constants, decay rates, or mixing angles—advancing the search for a unified geometric description of fundamental phenomena.

References

[1] Koide, Y. "A New View of Quark and Lepton Mass Hierarchy." Physics Letters B, vol. 120, no. 1-3, 1983.

[2] Daniel Banasik, *Wave Theory: A Unified Geometric Framework*. GitHub Repository: `https://github.com/WaveTheoryPhysics/WaveTheory`